Advanced Computer Systems Engineering Laboratory (ENCS5150)

TODO 3 – My Study Planner

---

1. Overview

Develop an Android application called My Study Planner for tracking student study sessions using three fragments and SharedPreferences. The application must save profile information, log study sessions, and display a summary using persistent data.

Note:

- The application must use three fragments: ProfileFragment, LogFragment, and SummaryFragment.
- The application must store and retrieve data using SharedPreferences through a singleton SharedPrefManager.

2. Application Structure

2.1 ProfileFragment

This is the initial screen of the application. The student enters their name, major, and daily study goal. The information is saved using SharedPreferences.

2.2 LogFragment

This screen allows the student to log a study session by entering the subject name and the number of hours studied.

2.3 SummaryFragment

This screen displays all saved profile and study data. It also includes a reset button to clear the saved SharedPreferences data.

3. Tasks to Build

1. Create the project and SharedPrefManager singleton

Create a new Android project and add a helper class for saving and reading app data. This class will be used to store the student's name, major, daily goal, total sessions, total hours, and subjects.

2. Build Profile Fragment

Create the first screen for setting up the student profile. The screen should have fields for the student name, major, and daily study goal. When the user clicks **Save Profile**, the data should be saved and a toast message should appear. The **Next** button should move the user to the study session fragment.

3. Build the Log Session Fragment

Create the second screen for adding a study session. The user should enter the subject name and the number of hours studied. When the user clicks Log Session, the app should save the subject name and update the total hours for this subject.

4. Build the Summary Fragment

Create the third screen to show the saved study information. It should display the student's name, major, daily goal, total subjects, total hours, and each subject with its hours. If there is no saved profile, show a message that no profile has been saved yet. Add a button to reset all saved data.

5. Wire up MainActivity with fragment navigation

Connect the three fragments so the user can move from the profile screen to the log session screen, then to the summary screen. When the app starts again, it should open the log session screen directly if the student profile was already saved.

Notes

- Set the **App Name** to: **Your Full Name-Your ID.**
- All SharedPreferences access must be done **only through SharedPrefManager**.
- Use **Pixel 3a XL device (Graphic = Software)**.
- Cheating or using LLM Models will result in **0 marks**.

Marks Distribution (Total: 10 Marks)

Component	Description	Marks
Functionality	Implementation of all required features; buttons, game logic, navigation, and correct handling of user actions	4
Error-Handling & Stability	Proper handling of invalid inputs, prevention of crashes, smooth app behavior under different scenarios	2
Design & User Interface	Clear, visually appealing layout, adherence to color schemes, hints, and layout requirements	2
Usability & User Experience	Ease of use, intuitive navigation, responsiveness, and consistency between activities	1
Documentation & Naming Conventions	Correct app name format (ID_FirstName_LastName), proper project structure, meaningful variable names, comments …	1

Submission Requirements

Submit it through Ritaj in the following manner

1. **Project.zip file** (Size in KB)
    - From Android Studio: File → Export → Export to Zip File
2. **app-debug.apk file** (Size in MB)
    - From Android Studio: Build → Build Bundle(s) / APK(s) → Build APK(s)
    - You will find the APK file under: app\build\outputs\apk\debug\app-debug.apk

Send both the APK file and the ZIP file as a reply to my message.